

India's Pearl & Coloured Gemstone Trade Intelligence Dashboard

Executive Summary

This project aims to develop an interactive **Gemstone Trade Intelligence Dashboard** in Power BI to analyse India's International Trade in **Natural Pearls, Cultured Pearls, Emeralds, Rubies and Sapphires**.

The project will use real, publicly available trade data from the **Government of India's Department of Commerce - TRADESTAT** to study import and export patterns across countries and over time.

Rather than simply reporting trade values, the project aims to turn raw trade statistics into meaningful business insights by identifying **important markets, emerging opportunities, sourcing patterns, market concentration and changes in trade behaviour**.

The proposed solution will combine data analytics, business intelligence and automated data processing to create a practical decision-support tool for understanding India's pearl and coloured gemstone trade.

Problem Statement

❖ Background

The gems and jewellery industry generates significant international trade, yet business analysis of the sector often focuses heavily on diamonds, gold and overall jewellery markets.

Pearls and coloured gemstones such as emeralds, rubies and sapphires are comparatively less explored from a data analytics perspective, despite having distinct sourcing markets, trading patterns and international demand.

India's official trade data contains detailed information about the movement of these commodities between India and countries around the world. However, this information is primarily available as individual trade statistics and requires substantial analysis to identify meaningful patterns.

Simply knowing that imports or exports have increased does not answer important business questions such as:

- Which countries are becoming important trading partners?
- Where is India's sourcing concentrated?
- Which markets are growing consistently?
- Are emerging markets being overlooked because their current trade volume is small?
- Which gemstone categories are showing stronger trade momentum?
- Is India's international trade becoming more diversified or more dependent on a few markets?

❖ Objective

To develop a Power BI-based trade intelligence solution that transforms official Indian trade statistics into **clear, actionable insights about the international trade of pearls and selected coloured gemstones**.

❖ Scope

The project will focus on five commodity categories:

- **Natural Pearls**
- **Cultured Pearls**
- **Emeralds**
- **Rubies**
- **Sapphires**

The analysis will cover **imports and exports, country-wise trade, historical trends, growth patterns and market-level comparisons**.

Data Sources

❖ Primary Data

The project will use official trade statistics published by the **Government of India, Ministry of Commerce and Industry, Department of Commerce – TRADESTAT**.

The data provides commodity-wise and country-wise information relating to India's imports and exports and will form the primary analytical dataset for the project.

❖ Supporting Data

Additional reference data may be used where required for:

- Country and geographic classification
- Commodity and HS-code classification
- Time-period standardisation
- Supporting analytical calculations.

The project will rely on **actual reported trade data rather than synthetic or simulated data**.

Methodology

1. Data Collection & Integration

Trade data for the selected commodities will be collected from the official government source and consolidated into a structured dataset covering imports and exports across relevant countries and time periods.

The data collection process will be designed so that the analytical dataset can be updated when new government trade information becomes available.

2. Data Cleaning & Transformation

The collected data will be cleaned and standardised to address:

- inconsistent country names,
- missing or inconsistent values,
- commodity classifications,
- time-period formats,
- duplicate records, and
- differences between import and export datasets.

The data will be maintained in a consistent analytical currency to enable meaningful comparison across commodities and countries.

3. Data Modelling & Analysis

The project will develop a structured data model to analyse relationships between:

- commodities,
- countries,
- time periods, and
- import/export activity.

Business and analytical measures will be developed to study trade value, growth, market share, trade balance and concentration.

Particular attention will be given to identifying **emerging markets and significant changes in sourcing and destination patterns**, rather than relying only on total trade rankings.

4. Business Intelligence & Visualisation

Power BI will be used to transform the processed data into an interactive business intelligence solution.

The dashboard will allow users to explore trade performance across commodities, countries and time periods and identify patterns that may be relevant to **market expansion, sourcing strategy and international trade decisions**.

Expected Outcomes

The project is expected to deliver:

- An interactive **Pearl & Coloured Gemstone Trade Intelligence Dashboard**.
- A consolidated dataset based on official government trade statistics.
- Automated and reusable data-processing workflows.

- Comparative analysis of Natural Pearls, Cultured Pearls, Emeralds, Rubies and Sapphires.
- Identification of major and emerging international markets.
- Insights into India's sourcing and export patterns.
- Analysis of market concentration and trade dependencies.
- A business-oriented framework for identifying potential opportunities and risks in international gemstone trade.

The ultimate aim is to move from **descriptive reporting to decision-oriented analysis**.

Tools & Technologies

- **Power BI Desktop** - dashboard development and interactive visualisation
 - **DAX** - analytical measures and KPI calculations
 - **Power Query** - data transformation and ETL
 - **Python** - automated data collection, processing and validation
 - **SQL** - structured data storage, querying and modelling
 - **Excel/CSV** - intermediate data handling where required
 - **Git/GitHub** - version control and project documentation
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Risks & Challenges

❖ Data Quality

Government datasets may contain missing values, revisions or inconsistencies that need to be identified and handled carefully.

❖ Classification Changes

Commodity and HS-code classifications may change across reporting periods, requiring appropriate validation before historical comparisons are made.

❖ Interpretation of Growth

Very high percentage growth may sometimes result from a small initial trade base. Growth therefore needs to be interpreted alongside market size and historical trends.

❖ Data Granularity

The available trade data primarily represents country-level trade and may not provide company-level or individual transaction information.

❖ Source Availability

The dashboard's refresh capability will depend on the availability and publication schedule of the underlying official government data.

Conclusion

GemStrategy aims to demonstrate how publicly available government data can be transformed into meaningful business intelligence for a specialised and relatively underexplored industry segment.

By focusing on **Natural Pearls, Cultured Pearls, Emeralds, Rubies and Sapphires**, the project moves beyond commonly developed generic dashboards and explores a more specialised business problem involving **international sourcing, market development and trade strategy**.

The project will combine real-world data with **Python, SQL and Power BI** to create an analytical solution that can identify not only what has happened in India's gemstone trade, but also **where meaningful changes, opportunities and potential dependencies are emerging**.

The intended outcome is therefore not simply a visual representation of trade statistics, but a **practical trade intelligence tool that demonstrates how data analytics can support real business decision-making**.